

The Exact Value $Z(12, 18; 3, 3) = 108$

An Exact Computation-Assisted Proof

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Abstract

We determine the finite Zarankiewicz number $Z(12, 18; 3, 3)$. An explicit 12×18 Boolean matrix with 108 ones supplies the lower bound. For the upper bound, two independently replayable orbit-certificate families establish $Z(12, 17; 3, 3) \leq 103$ and $Z(11, 18; 3, 3) \leq 101$. These deletion bounds force every hypothetical 109-one matrix to have column degrees $7^{16}1^7$ and minimum row degree at least eight. Four exhaustive cases remain, according to whether a degree-eight row exists and whether the unique degree-seven column contains the distinguished row. Each case is excluded by an exact symmetry-orbit certificate. A standard-library Python verifier reconstructs every orbit, inequality, rational dual calculation, and integer branch using exact arithmetic. It checks 354 neighboring degree patterns and all four final cases, and accepts the complete proof chain. Hence

$$\boxed{Z(12, 18; 3, 3) = 108}.$$

The complete witness, certificates, generators, mutation tests, hashes, and one-command replay package accompany this report. The result has not yet undergone independent peer review.

1 Introduction

For positive integers m, n, s, t , the Zarankiewicz number $Z(m, n; s, t)$ is the largest number of edges in a bipartite graph with parts of orders m and n that contains no $K_{s,t}$. In matrix language, $Z(m, n; 3, 3)$ is the maximum number of ones in an $m \times n$ Boolean matrix with no all-one 3×3 submatrix.

The small finite $K_{3,3}$ cases remain a useful benchmark for extremal graph theory and exact computation. Davies, Gill and Horsley developed strengthened linear-programming upper bounds for many small Zarankiewicz parameters [1]. Bhan, Nobili and Langer subsequently reported new lower bounds across the open $3, 3$ frontier and closed three cells, including $Z(12, 22; 3, 3) = 132$ [2]. Prior bounds were 108--113; this work proves equality at 108.

We write $[v] = \{1, \dots, v\}$ and identify every column with its support $B_j \subseteq [v]$. A family $(B_1, \dots, B_n)$ is admissible exactly when

$$\lambda_T := |\{j : T \subseteq B_j\}| \leq 2 \quad \text{for every } T \in \binom{[v]}{3}. \quad (1)$$

Repeated blocks are allowed.

Theorem 1. $Z(12, 18; 3, 3) = 108$.

2 The 108-one construction

The matrix in Appendix A has 108 ones. Every column has degree six, while the row degrees are

$$(10, 9, 9, 9, 9, 9, 7, 9, 8, 8, 10, 11).$$

Direct enumeration of the $\binom{12}{3} = 220$ row triples gives multiplicity histogram

$$6 \text{ triples of multiplicity } 0, \quad 68 \text{ of multiplicity } 1, \quad 146 \text{ of multiplicity } 2.$$

Thus (1) holds and

$$Z(12, 18; 3, 3) \geq 108. \quad (2)$$

3 Two self-contained deletion bounds

The final reduction uses two neighboring upper bounds, both proved inside the package rather than taken as external assumptions.

3.1 $Z(12, 17; 3, 3) \leq 103$

It is enough to exclude exactly 104 ones, because deleting ones preserves admissibility. Let $c_1 \geq \dots \geq c_{17}$ be the column degrees. Double counting pairs (T, j) with $T \subseteq B_j$ gives

$$\sum_{j=1}^{17} \binom{c_j}{3} \leq 2 \binom{12}{3} = 440. \quad (3)$$

The exact verifier independently enumerates every nonincreasing integer 17-tuple with sum 104 satisfying (3). There are 303. The 303 compressed certificates exclude all of them.

3.2 $Z(11, 18; 3, 3) \leq 101$

Similarly, a 102-one 11×18 matrix would have a nonincreasing degree sequence satisfying

$$\sum_{j=1}^{18} c_j = 102, \quad \sum_{j=1}^{18} \binom{c_j}{3} \leq 2 \binom{11}{3} = 330.$$

There are exactly 51 such sequences, and the package contains an accepted certificate for each. Therefore

$$Z(11, 18; 3, 3) \leq 101. \quad (4)$$

4 Reduction of 109 ones to four cases

Suppose for contradiction that an admissible 12×18 matrix has 109 ones.

If a column has degree at most five, deleting that column leaves at least $109 - 5 = 104$ ones in a 12×17 matrix, contradicting the preceding bound. Hence every column has degree at least six. Their sum is 109, so the multiset is forced:

$$7^1 6^{17}. \quad (5)$$

If a row has degree at most seven, deleting that row leaves at least $109 - 7 = 102$ ones in an 11×18 matrix, contradicting (4). Thus every row has degree at least eight.

If a degree-eight row exists, normalize one. The unique degree-seven column either contains it (case A) or avoids it (case B). If no degree-eight row exists, every row has degree at least nine; the row-degree sum 109 then forces

$$10^1 9^{11}. \quad (6)$$

The unique degree-seven column either contains the unique degree-ten row (case I) or avoids it (case O). These four cases are exhaustive.

5 Symmetry-orbit certificates

We describe the common exact certificate language.

5.1 Row cells and block orbits

After fixing some blocks, two rows are equivalent when they have the same membership signature in all fixed blocks. Let the resulting cells have sizes $n_1, \dots, n_s$. The stabilizer of the fixed structure is

$$H = \text{Sym}(n_1) \times \dots \times \text{Sym}(n_s).$$

An orbit of candidate k -blocks is represented by a profile $p = (p_1, \dots, p_s)$ with $0 \leq p_i \leq n_i$ and $\sum_i p_i = k$. Its size is

$$N_p = \prod_i \binom{n_i}{p_i}. \quad (7)$$

Likewise, a triple orbit is represented by $u = (u_1, \dots, u_s)$ with $\sum_i u_i = 3$ and has size

$$N_u = \prod_i \binom{n_i}{u_i}.$$

A block of profile p contains

$$h(u, p) = \prod_i \binom{p_i}{u_i} \quad (8)$$

triples of profile u .

If fixed blocks already cover every triple in orbit u exactly λ_u times, and x_p is the number of remaining blocks selected from orbit p , then every completion satisfies

$$\sum_p h(u, p) x_p \leq (2 - \lambda_u) N_u. \quad (9)$$

Exact block-count and multiplicity upper bounds are appended.

5.2 Marked pair capacities in cases A and B

Delete the normalized degree-eight row. The eight columns incident with it are marked. Any pair of the remaining eleven rows can occur in at most two marked blocks; otherwise that pair together with the deleted row lies in three columns. The verifier therefore constructs pair orbits and inequalities parallel to (9). It also imposes aggregate residual-row degree demands.

The residual type counts are:

Case	Fixed block	Remaining block types
A	marked size 6	7 marked size 5; 10 unmarked size 6
B	unmarked size 7	8 marked size 5; 9 unmarked size 6

5.3 Exact dual leaves

At every node the verifier reconstructs an integer system

$$Ax \leq b, \quad l \leq x \leq u.$$

For any nonnegative rational vector y , put $z = x - l$. Then

$$\mathbf{1}^T x = \mathbf{1}^T l + y^T A z + \sum_j (1 - (A^T y)_j) z_j \quad (10)$$

$$\leq \mathbf{1}^T l + y^T (b - A l) + \sum_j (u_j - l_j) \max\{0, 1 - (A^T y)_j\}. \quad (11)$$

A dual leaf stores y as nonnegative integer numerators over one positive common denominator. Equation (11) is evaluated with integers only. If the resulting upper bound is strictly below the required number of remaining columns, the node is closed.

An inner split branches exhaustively on $x_j \leq q$ versus $x_j \geq q + 1$. An outer split orders candidate H -orbits, takes the least occupied orbit, fixes its canonical representative, and forbids all earlier orbits. The forbidden union is H -invariant; hence every completion lies in exactly one child. Induction proves exhaustiveness.

6 Verified certificate totals

Table 1 records the deterministic counts recomputed by the verifier.

Component	Patterns	Outer	Inner	Dual	Integer
$Z(12, 17) \leq 103$	303	3,607	26,725	14,651	11,764
$Z(11, 18) \leq 101$	51	160	785	451	320
Case I	1	1	1	1	0
Case O	1	71	217	139	75
Case A	1	77	6,290	3,077	3,109
Case B	1	31	2,696	1,342	1,335

Table 1: Exact replay totals. “Dual” denotes rational dual leaves and “Integer” exhaustive integer splits.

All four final certificates are accepted. Consequently a 109-one matrix does not exist. Together with (2), this proves Theorem 1.

7 Reproducibility and trust boundary

The trusted base consists of the Python standard library, the standalone verifier, and the hashed witness/certificate files. The verifier does not import the generators and does not call an LP, MILP

or SAT solver. It independently regenerates degree patterns, row cells, candidate orbits, capacities and all branch semantics. Rational leaves are checked using Python integers.

The one-command gate checks SHA-256 hashes, replays the proof, compares deterministic report invariants, and runs three adversarial mutation tests. The mutations alter the witness, a rational leaf and bridge metadata; all are rejected.

The included generator scripts used NumPy and SciPy/HiGHS to discover certificates. They are not load-bearing. The result should be regarded as a new computer-assisted proof pending independent reproduction and peer review.

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A The 108-one witness

The rows of the matrix are:

```

0 1 1 0 1 0 1 1 1 0 1 1 0 0 0 1 0 1
1 0 1 1 0 0 1 1 1 0 0 1 1 0 1 0 0 0
0 1 1 0 0 1 1 0 0 1 1 0 1 0 1 0 0 1
0 1 1 1 1 0 0 0 0 1 0 1 0 1 1 1 0 0
0 1 0 1 0 1 1 0 1 1 0 0 1 0 0 1 1 0
0 1 0 0 1 1 0 1 1 0 1 0 0 1 1 0 1 0
1 0 0 0 1 0 1 0 0 1 1 1 0 0 0 0 1 0
1 0 1 1 1 1 0 0 1 1 0 0 0 1 0 0 0 1
1 0 1 0 0 1 0 1 0 0 1 0 1 1 0 1 0 0
0 1 0 1 0 0 0 1 0 0 0 1 1 1 0 0 1 1
1 0 0 1 1 1 1 1 0 0 0 0 0 0 1 1 1 1
1 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1

```

Equivalently, the column supports are:

Column	Support
1	{2, 7, 8, 9, 11, 12}
2	{1, 3, 4, 5, 6, 10}
3	{1, 2, 3, 4, 8, 9}
4	{2, 4, 5, 8, 10, 11}
5	{1, 4, 6, 7, 8, 11}
6	{3, 5, 6, 8, 9, 11}
7	{1, 2, 3, 5, 7, 11}
8	{1, 2, 6, 9, 10, 11}
9	{1, 2, 5, 6, 8, 12}
10	{3, 4, 5, 7, 8, 12}
11	{1, 3, 6, 7, 9, 12}
12	{1, 2, 4, 7, 10, 12}

Column	Support
13	$\{2, 3, 5, 9, 10, 12\}$
14	$\{4, 6, 8, 9, 10, 12\}$
15	$\{2, 3, 4, 6, 11, 12\}$
16	$\{1, 4, 5, 9, 11, 12\}$
17	$\{5, 6, 7, 10, 11, 12\}$
18	$\{1, 3, 8, 10, 11, 12\}$

B Package replay

From a clean extraction, run

```
./verify.sh
```

The verifier writes a fresh report and prints

```
VERIFIED: Z(12,18,3,3)=108
```

only after all 354 neighboring degree patterns and the four final cases have been accepted.

References

- [1] S. Davies, P. Gill and D. Horsley, *Improved upper bounds on Zarankiewicz numbers*, Discrete Mathematics 349(5) (2026), Article 114924. DOI: 10.1016/j.disc.2025.114924.
- [2] J. Bhan, N. Nobili and P. Langer, *New Bounds for Zarankiewicz Numbers via Reinforced LLM Evolutionary Search*, arXiv:2605.01120 (2026).